Bryce Forsgren

Cambridge, MA | bfors15@mit.edu | linkedin.com/in/bryce-forsgren

EDUCATION

Massachusetts Institute of Technology (MIT) Cambridge, MA
B.S. Mechanical Engineering & B.S. Applied Mathematics; Minor in Spanish | GPA: 4.9/5.0 *June 2027*
Coursework: Advanced Heat & Mass Transfer; Fluid Mechanics; Aircraft Engines & Gas Turbines; Mechanics & Materials II; Design & Manufacturing II; Measurement & Instrumentation; Design Thinking for Innovation, Leadership, & Engineering
Honors: Pi Tau Sigma ME Honor Society (2025);
International Design Competition Award (competed against ∼130 students and placed fifth) (2025)

EXPERIENCE

SpaceX Cape Canaveral, FL
Dragon Final Integration Launch Engineering Intern *June 2026 – Aug 2026*
- Built Excel VBA tool utilizing SQL queries to pull every open ticket blocking mission-critical Dragon closeouts; cut a daily 15–30 min task down to <30 s; deployed to all Dragon departments (∼100 people) for all future missions, saving 30+ hrs/mission
- Designed thruster cluster protective covers from scratch in Siemens NX, preventing impact damage that could cost ∼200 labor hrs to remove & replace; ran material (abrasion and adhesive) and fitment tests on a development capsule; led design reviews with ∼20 engineers, which included propulsion leads
- Prototyped a Python tool that auto-imports 1,000+ fasteners into active mission-critical work orders via a remote Linux worker; projected ∼30 hrs/mission saved
- Supported 30+ technicians through final integration of Dragon for NASA Crew-13, CRS-36, & NASA Crew-14

MIT Motorsports (Formula SAE Electric) Cambridge, MA
Testing Lead *Aug 2025 – Present*
- Led cross-country test campaign at Ford's wind tunnel and Toyota Racing Development for the team's first-ever aero map and K&C data
- Organized 40 test trips (∼400 km on car) for the ∼150-member team, leading maintenance and scrutineering to keep the driver safe
- Instrumented push-rod strain gauges and tire/brake temperature sensors; involved in inverter/battery faults and debugging

Composites Sub-Team Analyst *Sep 2024 – Present*
- Developed a CLPT study via MATLAB of 2x2 twill carbon fabric via tensile and 3-point bend analysis on a mechanical tester, as well as microscopy of failure modes; found a constant correction factor for up to 89% error
- Authored 35 page paper on the CLPT findings for woven composites (in preparation for publication)
- Owned rear-wing build through endplates and bonded joints under tight timeline 3 days before the 2026 competition
- Wet laid up 7 aerodynamic body panels in under 2 weeks before the 2025 competition; included post-processing and weight optimization

RESEARCH

MIT BioInstrumentation Lab (UROP) Cambridge, MA
Undergraduate Researcher, Fonterra Membrane Industry-Sponsored Project *June 2024 – June 2026*
- Supported construction of high-pressure membrane test cart under tight timelines before 3 sponsor meetings with Fonterra
- Designed and soldered an I^2C master–slave perf board linking 6 high-flow/low-flow sensors
- Measured high and low flow across 3 parallel paths at constant flux, which ran robustly through 10+ tests

Undergraduate Researcher, Precision Flexure Clamps *Mar 2024 – Dec 2024*
- Designed 30 in-plane flexure clamps for optics tables and test stands; validated by analytical models and FEA
- Made 15 working clamps with repeatable, non-yielding clamping from 3 machine shops; co-inventor on accepted MIT TLO disclosure
- Presented research poster at the 39th ASPE Annual Meeting to leading precision engineering researchers

Undergraduate Researcher, Fluidic Optical Shaping *Dec 2024 – June 2026*
- Built Mathematica PDE optimization lens-shaping model; COMSOL wetting sims identified poor surface wetting as fabrication barrier

TEACHING & MENTORSHIP

Undergraduate TA, 2.006 Thermal-Fluids Eng. II – office hours and tutoring for 60 students *Feb 2026 – May 2026*
MechE Department Tutor – mechanics, dynamics, controls, thermal-fluids, electronics, etc. for ∼100 peers *Sep 2025 – Present*
MechE Private Tutor – Dyanamics and Controls II support for visually impaired student *Feb 2026 – May 2026*
SHED Lab Mentor (MIT EHS) – maintained advanced fabrication lab & oversaw research projects *Feb 2025 – May 2026*

SKILLS

CAD/CAE: Siemens NX, SolidWorks, Autodesk Fusion, Onshape; ANSYS, COMSOL
Programming: Python, MATLAB, Mathematica, SQL, Excel VBA, ROS2, Linux, LaTeX
Fabrication: Composite wet layup & bonding, CNC machining & Fusion CAM, wire EDM, waterjet, laser cutting, FDM printing